PRODUCT REQUIREMENT DOCUMENT (PRD)

FOR TEAM BAKKUTTEH

UMHackathon 2026

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Project Name: Crypto-Integrated Financial Intelligence Advisor (CIFA)

1. Project Overview

- **Problem Statement:** In Malaysia and the rest of the world, many individuals, businesses, and communities face challenges in making timely and informed decisions due to fragmented data, limited analytical capabilities, and reliance on manual or intuition-based processes.
- **Target Domain:** AI for Economic Empowerment & Decision Intelligence
- **Proposed Solution Summary:** Our CIFA is an AI-powered financial decision intelligence platform powered by Z.AI's GLM that solves Malaysia's fragmented financial data problem by aggregating EPF, FD rates, and Bitcoin data into a unified dashboard, delivering personalized investment recommendations in minutes rather than hours of manual research. CIFA aims to provide context-aware, and explainable recommendations tailored to each user's unique situation and the current market data to transform financial illiteracy to simplicity and clarity

2. Background & Business Objective

- **Background of the Problem:** For years, financial planning in Malaysia has been a fragmented, manual process. Individuals must navigate multiple siloed platforms (EPF i-Akaun, various banking apps/websites for up-to-date FD rates, and crypto exchanges) and manually cross-reference this data against static age-based benchmarks. This requires high financial literacy, hours of research, and often leads to emotional, uninformed decision-making.

- **Importance of Solving This Issue:** Consolidating this fragmented data into a single, high-clarity interface is critical because it replaces manual research with AI-driven research and reasoning. Thus, regardless of financial literacy, working Malaysians with

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little time to spare can make well-informed decisions and never miss high-yield opportunities.

- **Strategic Fit / Impact:** CIFA aligns with the Z.AI Economic Empowerment Ecosystem. By integrating the Z.AI GLM, we transform live crypto market data, fixed deposit rates, and EPF rates into a dynamic "Decision Intelligence" workflow. Therefore, directly supporting Malaysia's goal for financial stability and retirement adequacy.

3. Product Purpose

3.1. Main Goal of the System: To provide AI-powered financial reasoning based on the user's unique situations and current market data.

3.2. Intended Users (Target Audience):

- Working-class Malaysians
- Students
- Freelancers

4. System Functionalities

4.1. Description: CIFA operates as a high-speed financial intelligence engine that transforms fragmented financial data - EPF standards, fixed deposit Rates and live cryptocurrency data, into personalized recommendations through natural language interaction. Users input their financial profile (age, monthly salary and expenses, current EPF, FD, and Crypto balance and their EPF saving category) and receive multi-asset recommendations (EPF, FD, and Bitcoin) with projected allocated surplus.

4.2. Key Functionalities:

- **Vibe Analysis and Health Scoring:** Translates financial profiles into health metrics (EPF Health Score 0-100%, Status: Behind, On Track, and Way Behind).
- **Conversational Intelligence:** Facilitates continuous changes to the user profiles, updating the reasoning and explaining why each allocation is prioritized.

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- **Risk Modulation:** Caps high-risk assets (Crypto) regardless of user's surplus due to its high volatility
- **Threshold-Aware Allocation:** Intelligently pivots funds to EPF if bank FD minimums (RM 500) are not met.
- **Real-Time Data Integration:** Uses real-time bitcoin cryptocurrency rates, suggesting to the user when to buy, hold or sell

4.3. AI Model & Prompt Design

4.3.1. Model Selection: We chose Z.AI GLM due to its high-level reasoning, which is critical for financial clarity of our program and it understands the Malaysian context.

4.3.2. Prompting Strategy: Our team uses a multi-step agentic prompting: where we first parse user inputs, extract the structured data, analyze the EPF status, evaluate liquidity, assess the crypto market, generate surplus allocation and provide reasoning for the allocation. Multi-step agentic prompting fits our use case because it breaks down the complex reasoning into digestible chunks and allows for each step to validate against financial rules.

4.3.3. Context & Input Handling: CIFA utilizes a layered sanitization pipeline to ensure that oversized or unstructured data does not lead to "decision hallucination". The maximum input for our inputs are age (between 18 and 120), monthly salary and expenses (between RM 0 and RM 1,000,000), and other balances are (between RM 0 and RM 20,000,000).

4.3.4. Fallback & Failure Behavior: CIFA uses a 3-tier failure and fallback strategy in which the primary tier involves the Z.AI GLM providing full reasoning and personalized recommendations. However, if Z.AI fails, the request is rerouted to Genimi AI to fill in Z.AI's place. Finally the last tier involves a fixed fallback logic with short unpersonalized explanations if both previous tiers fail.

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5. User Stories & Use Cases

- **User Stories:** As a fresh graduate with RM 1,000 monthly surplus, I want to get personalized investment recommendations that explain the trade-offs so that I feel confident in my choices and never miss crucial financial opportunities.

- **Use Cases (Main Interactions):**

  Prompt to UI Flow: Users enter their financial profile, CIFA audits the EPF status and evaluates liquidity. Following this, Z.AI generates a percentage-based breakdown with reasoning.

  Flow Generation: Users update their financial profile, then CIFA re-evaluates the allocation while validating financial rules and ensuring that EPF is prioritized and cryptocurrency is capped.

- **Use Case Diagram:** *(Optional but encouraged for participants to visualize the interaction)*.

6. Features Included (Scope Definition)

- Unified Financial Dashboard, aggregation EPF, FD rates and live Bitcoin cryptocurrency rates
- Real-Time Data Integration (Bitcoin cryptocurrency)
- FD Comparison and Ranking, displays the banks with the highest FD rates and their minimum amounts
- Visual EPF Health Scoring, use colour-coded charts to show if users are behind or on track with their chosen EPF category (basic, adequate or enhanced)

7. Features Not Included (Scope Control)

- User Authentication
- Stock Markets

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- Multiple fixed deposit inputs
- Multiple cryptocurrencies

8. Assumptions & Constraints

- **LLM Cost Constraint:** Based on the current system token usage, the average user session consumes approximately 2,020 tokens per request. This is calculated from a cumulative total of 252,359 tokens across 125 requests. To keep costs manageable at scale, CIFA utilizes a "Compute-First, Reason-Second" architecture. By offloading complex financial calculations (deficit percentage, priority leveling, and age-standard lookups) to the local python EPF Calculator, we avoid the high token overhead required for "Chain-of-Thought" prompting that would otherwise be needed for an LLM to perform these mathematical operations accurately.
- **Technical Constraints:** Due to legal issues, current FD Rates were written manually and fed to CIFA instead of daily updated via bank APIs.
- **User Input:** CIFA is unable to generate allocations or reasoning without user financial profile inputs and is not ready for production so requires human approval.

9. Risks & Questions Throughout Development

- **Financial Accuracy:** How can we ensure that FD rates, EPF standards and cryptocurrency are up-to-date and always accurate?
- **Z.AI GLM Poor Financial Reasoning:** What if the Z.AI GLM hallucinates and provides false financial advice
- **User Personalization**: How can we ensure that each user requires a unique allocation based on their specific financial profile.

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